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End Semester Examination Dec. 2025

Name of the Course: B.Tech (ECE-ESR)

Semester: V

Name of the Paper: Sensor Technology

Paper Code: **TEC 551**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
	Describe the following terms:	
(a)	i. Span. ii. Accuracy iii. Transfer function	CO1
(b)	Explain dead band and repeatability with proper diagram.	
(c)	Explain the term calibration and non-linearity with respect to sensors.	
Q2	(20 marks)	
(a)	Explain Hall-effect, and describe its application in detail.	
(b)	Write a short note on following terms: i. Piezoelectric effect ii. Thermoelectric effect	CO2
(c)	Describe the dynamic models of sensor elements.	
Q3	(20 marks)	
(a)	Describe about Interface circuits of sensor.	CO3
(b)	Describe charge amplifiers in detail with proper diagrams.	
(c)	Explain voltage to current converter with proper circuit diagrams.	
Q4	(20 marks)	
(a)	Describe the working of LVDT in detail.	CO4
(b)	Describe ultrasonic detectors, how it can be used in position measurement.	
(c)	Explain in detail the working of RVDT with proper diagram.	
Q5	(20 marks)	
(a)	Describe the concept of resistive and piezoelectric humidity sensors.	CO5
(b)	What are image sensors. Explain CCD imaging sensor in detail.	
(c)	Describe gas flame detectors.	

